

HASI-Net: A Heterogeneous Adaptive Spatio-Temporal Informer Network for Forecasting Aggregated Crime Counts

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Abstract—Forecasting crime from aggregated regional records is harder than forecasting from precise incident coordinates: the spatial unit is coarse, the series are short, the counts are zero-inflated and right-skewed, and the inter-region graph is only partially observed. We make two contributions. First, we propose HASI-Net, a Heterogeneous Adaptive Spatio-temporal Informer network whose graph block fuses geographic rook contiguity, socioeconomic k -NN cosine similarity, and a learnable latent adjacency through learned softmax weights; whose multi-scale temporal block combines a ProbSparse Informer encoder, a dilated TCN, and a series-decomposition trend/seasonal split behind a learned gate; and whose persistence-residual count-aware head predicts a per-horizon delta on a lookback-mean carry, starting at the historical-average baseline and emitting a zero-inflated negative-binomial parametrisation. Second, and equally important, we report a rigorous eight-model, five-seed, three-horizon benchmark on a real-geography Chicago community-area $\times$ month panel (120 months, 77 areas, four unified women-related crime categories), evaluated not only on point error but on a leak-free event-verification protocol (per-node 90th-percentile spike thresholds from training only; POD/FAR/CSI and top- k hotspot hit rate). The central, honestly-reported finding is that simple historical-average persistence is a remarkably strong baseline—it leads raw MAE at every horizon and is matched, not beaten, by every deep model including HASI-Net on both point and event metrics. GraphWaveNet is the marginal accuracy winner; HASI-Net is the best deep model at the medium horizon and the most reproducible at the long horizon, and a component ablation confirms each graph channel contributes. We argue the deep model’s value therefore lies not in point accuracy but in the calibrated probabilistic forecasts and cross-region transfer developed in the companion paper. All numbers come from re-runnable CUDA runs with persisted predictions.

Index Terms—spatio-temporal forecasting, graph neu-

ral networks, Informer, crime prediction, adaptive graph learning, count-aware loss, forecast verification, persistence baseline

I. Introduction

Crime prediction uses historical records to forecast future incident counts across regions and time. With digitised records, deep models can surface patterns that elude manual analysis [1], [2]. However, most established crime-forecasting models assume fine-grained spatial information—exact latitude and longitude per incident—which is available for cities such as Chicago but not for official Indian data from the National Crime Records Bureau (NCRB), which is published aggregated at the district/year level [3]. Designing a model that works on such coarse, short, zero-inflated aggregated data while still capturing meaningful spatio-temporal structure is the open problem this paper addresses.

A second, less advertised difficulty is that the regime itself penalises ambition. Monthly regional crime counts are highly persistent: a region’s near-future count is well approximated by its recent mean, so a trivial historical-average carry is a notoriously hard baseline to beat on the level, a fact documented across traffic, electricity and crime forecasting [4], [5]. Many spatio-temporal crime studies report a single seed and a narrow metric set (point MAE/RMSE), which quietly inflates the apparent advantage of complex models and obscures whether the gain is real or an optimisation artefact. A central methodological commitment of this paper is therefore not to claim a point-accuracy state of the art, but to measure the deep-model edge—if any—against a proper persistence baseline under multi-seed, multi-horizon,

leak-free event-verification, and to report the answer honestly even when it is unflattering to the proposed model.

This paper is one of two companions. The present paper develops the methodology and the point/event benchmark: the HASI-Net architecture, the heterogeneous adaptive graph, the multi-scale temporal block, the persistence-residual head, and the eight-model Chicago benchmark that establishes the strong-persistence finding. The companion paper builds on that finding directly: if point accuracy is saturated by persistence, the deep model must justify itself probabilistically, so the companion develops a sparsity-aware calibrated count head, conditional-coverage analysis on the high-crime tail, and cross-region transfer (Chicago→Austin, Chicago→Madhya Pradesh). The two papers are intentionally separable: the benchmark and the architecture stand on their own here, and the probabilistic and transfer contributions stand on their own there.

Recent hybrid models combine an attention-based temporal learner (Informer) with a graph convolutional spatial learner (ST-GCN) and outperform recurrent baselines on fine-grained Chicago data [2]. Adaptive graph methods learn the adjacency from data rather than fixing it [6], [7], [8], and multi-type relation graphs with focal loss handle crime-type imbalance [9]. Yet these methods are evaluated on precise-coordinate datasets and do not address the aggregated, short-series regime that dominates Indian regional crime reporting.

Contributions. (1) A heterogeneous adaptive graph block that fuses a fixed geographic prior, a fixed socioeconomic prior, and a learnable latent adjacency through learned softmax weights (Sec. III-B). (2) A multi-scale temporal block—series decomposition, ProbSparse Informer, dilated TCN, learned gate—tuned for short aggregated series (Sec. III-C). (3) A persistence-residual count-aware head with per-step deltas and a zero-inflated negative-binomial parametrisation, which anchors the deep model at the historical-average baseline and emits calibrated probabilistic forecasts (Sec. III-D). (4) A rigorous eight-model, five-seed, three-horizon benchmark on a real-geography Chicago panel, evaluated with point metrics, a leak-free per-node event-verification protocol (POD/FAR/CSI), and top- k hotspot hit rate—and the honest finding that historical-average persistence is a strong baseline that the deep models match but do not beat on point or event accuracy (Sec. VI).

(5) A component ablation isolating the contribution of each graph channel (Sec. VI-B). The calibrated probabilistic head, cross-region transfer, and conditional-coverage analysis that justify the deep model where point accuracy cannot be developed in the companion paper. All results are reproduced by open CUDA notebooks with persisted predictions.

II. Related Work

Spatio-temporal crime forecasting. Early neural approaches model crime as a sequence of city-wide counts or a per-region time series: Mao et al. [10] couple a CNN spatial encoder with an LSTM temporal decoder, and Deng [1] surveys risk-based deep predictors. Fan et al. [2] integrate Informer with ST-GCN for four Chicago crime types, showing that attention-based temporal learning combined with graph convolution beats recurrent networks; their setting, like most subsequent work, relies on precise incident coordinates. Shan et al. [8] (Ada-GCNLSTM) and Qin et al. [7] (ACSAformer) introduce adaptive graph construction and sparse attention for fine-grained urban forecasting. Wang et al. [9] (MRAGNN) model multi-type crime correlations with focal loss. Sui et al. [11] use graph attention for street-level incidents, and Hakyemez and Badur [12] augment street-network graphs with activity indicators. Surveys [13] and non-Western studies [14] confirm that inter-regional spatial correlations materially help regional crime forecasting. A consistent gap across this body of work is evaluation under precise-coordinate data with point metrics and limited seeds; none targets the aggregated, short-series Indian district regime, and few report a persistence baseline at all.

Graph representation learning. Graph convolution networks [15] propagate features over a fixed adjacency and are the spatial primitive underlying ST-GCN [16]. Graph WaveNet [6] introduces a learnable adaptive adjacency that does not require a predefined graph, a capability we generalise to heterogeneous priors. DCRNN and MTGNN extend graph propagation with diffusion convolutions and adaptive adjacency respectively; Informer-only isolates the temporal attention pathway. Our fusion of multiple adjacency notions through learned softmax weights is the architectural contribution relative to this line.

Time-series backbones. Informer [17] introduces ProbSparse self-attention for long-horizon efficiency; ST-GCN [16] and Graph WaveNet [6] establish graph convolution and adaptive adjacency for spatial-

temporal forecasting. DLinear [4] shows that a moving-average series decomposition is a strong, simple baseline on short series, motivating our decomposition step. TCN [5] and N-BEATS [18] inform the local temporal branch and the residual forecasting head respectively. The decomposition-plus-gated-branches design we adopt is a deliberate response to DLinear’s evidence that, on short series, a moving average is hard to beat with heavy attention—we keep the moving average as an explicit trend branch and let attention and the TCN compete only over the residual.

Count-aware losses and optimisation. Crime counts are zero-inflated and right-skewed; the zero-inflated negative binomial [19], [20], [21] and focal reweighting [22] address this for imbalanced count regression. We adopt a log1p-MSE default with a ZINB variant for ablation, and discuss in Sec. VI-B why a pure count likelihood mis-centres the quantiles on these small trended panels. For hyperparameter selection we use particle swarm optimisation [23] with linearly decreasing inertia [24] in a multi-swarm configuration [25].

Forecast evaluation. Comparing forecasters over multiple series and seeds requires non-parametric ranks: Friedman’s test [26] and paired comparisons with bootstrap confidence intervals [27] are the standard machinery for deciding whether an observed ordering is real [28]. We apply them at each horizon against the HA baseline rather than reporting a single-seed point table, and we complement point error with the leak-free event-verification protocol of Sec. V—POD/FAR/CSI originate in meteorological forecast verification and are the operationally relevant axis for policing.

III. Methodology

A. Problem formulation

Let $X \in \mathbb{R}_+^{T \times N \times C}$ be a panel of non-negative crime counts over T time steps, N spatial units (districts or community areas) and C crime categories, with node features $F \in \mathbb{R}^{N \times d_f}$ (census socioeconomics; identity where unavailable). Given a lookback window T_w and forecast horizon H , the model maps $X_{t:t+T_w} \in \mathbb{R}_+^{T_w \times N \times C}$ to a distribution over $X_{t+T_w:t+T_w+H} \in \mathbb{R}_+^{H \times N \times C}$. We predict the parameters of a zero-inflated negative binomial per (n, c, h) : log-mean $\log \mu$, log-dispersion $\log \alpha$, and zero-inflation logit π . The default

point-training objective is the log1p mean-squared error

$$\mathcal{L}_{\log 1p} = \frac{1}{NHC} \sum_{n,c,h} (\log 1p(y) - \log 1p(\hat{\mu}))^2, \quad (1)$$

which is scale-robust, well-behaved at zero, and—unlike a pure count likelihood—does not pull the mean off the persistence carry; the count-aware variant replaces it with the ZINB negative log-likelihood of Eq. (10) plus focal reweighting [22]. The hyperparameters, training schedule, and PSO budget are given in Sec. III-F.

B. Heterogeneous adaptive graph block

Three adjacency channels encode distinct, complementary notions of inter-region relatedness:

$$A_{\text{geo}} = \text{rook-contiguity, row-normalised}, \quad (2)$$

$$A_{\text{socio}} = k\text{-NN cosine on census features}, \quad (3)$$

$$A_{\text{adapt}} = \text{softmax}(\text{ReLU}(EE^\top)), \quad (4)$$

where $E \in \mathbb{R}^{N \times d_e}$ is a learnable node embedding. The three channels are constructed to be complementary, not redundant: A_{geo} encodes physical proximity and is constant across time; A_{socio} encodes demographic similarity that is stable but coarser than physical adjacency and, on the Chicago panel where census features are unavailable, reduces to a near-uniform global-smoothing prior (the socio channel is therefore exercised with genuine census features only on the MP panel, where Indian Census socioeconomics exist); A_{adapt} has no fixed prior at all and is free to discover latent interaction structure (e.g. crime-correlated commuting or policing catchments) that neither fixed channel captures. Row-normalising A_{geo} and A_{socio} keeps the per-node in-degree constant so that graph convolution does not favour densely-connected nodes; the softmax over logits α further guarantees the fused A is a valid convex combination whose row sums are bounded, stabilising training. The fused adjacency is

$$A = \text{softmax}(\alpha)_1 A_{\text{geo}} + \text{softmax}(\alpha)_2 A_{\text{socio}} + \text{softmax}(\alpha)_3 A_{\text{adapt}}, \quad (5)$$

with learnable logits $\alpha \in \mathbb{R}^3$. This lets the model down-weight a noisy prior (e.g. a coarse socioeconomic proxy) when the latent channel is more informative—the key adaptation for aggregated data where fixed priors are weak. Graph convolution proceeds as

$$H' = \text{LayerNorm}(\text{Dropout}(\text{GELU}(AHW_\ell)) + H), \quad (6)$$

stacked over L_g layers (Algorithm 1). The residual connection, GELU non-linearity, dropout and Layer-Norm follow standard practice [15]; the residual is important here because, with a strong persistence prior, the graph block should refine the carry rather than replace it, and a skip connection lets the optimiser learn a near-identity map when the graph signal is weak. The ablation in Sec. VI-B isolates each channel.

C. Multi-scale temporal block

A DLinear-style moving-average series decomposition splits each node’s hidden sequence into trend and seasonal components:

$$\text{trend} = \text{AvgPool}_k(H), \quad \text{seasonal} = H - \text{trend}, \quad (7)$$

with edge-replicated padding so the trend length equals the input length for any kernel k . The seasonal component is processed by two parallel branches: a ProbSparse Informer encoder [17] for long-range dependencies and a dilated causal TCN [5] for local dynamics. ProbSparse attention scores queries by the sparsity metric $M(q_i) = \max_j \tilde{K}_j q_i^\top / \sqrt{d} - \text{mean}_j(\cdot)$ and attends only with the top- Q queries; for our short sequences ($T_w \leq 32$) it degenerates safely to full attention, so the Informer branch contributes expressivity (multi-head global mixing of the seasonal residual) rather than computational efficiency on this benchmark. The dilated causal TCN stacks residual blocks with exponentially growing dilation $\{1, 2, 4, \dots\}$ so its receptive field covers the full look-back at modest depth while preserving causality—it captures the local month-to-month autocorrelation that the global attention branch may over-smooth. A learned scalar gate fuses the branches:

$$Z = \text{trend} + \sigma(g) \text{Informer}(\text{seasonal}) + (1 - \sigma(g)) \text{TCN}(\text{seasonal}), \quad (8)$$

with g initialised to 0 ($\sigma = 0.5$, an even mix). Algorithm 2 summarises the block.

D. Persistence-residual count-aware head

Aggregated crime counts are highly persistent: a strong, hard-to-beat baseline is the historical average (HA), i.e. the lookback mean carried forward flat across the horizon. We therefore decode residually: the head predicts a per-horizon-step delta on top of an encoded carry, so the model starts exactly at the HA

baseline (small-initialised head, $\Delta \approx 0$) and training only improves from there:

$$\begin{aligned} \text{carry} &= \frac{1}{T_w} \sum_t X_t, \\ \log \mu_h &= \text{enc}(\text{carry}) + \Delta \mu_h, \quad h = 1 \dots H, \end{aligned} \quad (9)$$

where $\text{enc}(\cdot) \in \{\log, \log 1p, \text{id}\}$ depends on the loss parametrisation. The per-step delta $\Delta \mu_h$ (rather than one flat delta) lets the model learn a trend or seasonal shape across the horizon—the structural advantage over flat HA, and the only mechanism by which a deep model can beat persistence on a persistent series. The head is small-initialised so that at the start of training $\Delta \mu_h \approx 0$ and the forecast collapses to the HA carry; gradients then only need to move the model away from an already-reasonable baseline, which is far easier than learning the level from scratch and avoids the Transformer failure mode of drifting off a strong carry on short series. Alongside $\log \mu$ the head emits $\log \alpha$ (NB dispersion) and π (zero-inflation logit) for calibrated probabilistic forecasts (Algorithm 3). The zero-inflation logit is warm-started from each node’s empirical zero fraction so that the gate reflects the data’s baseline sparsity rather than being learned from scratch; this is the count-aware variant (Sec. VI-B) and the parametrisation the companion paper builds its calibrated quantile head upon. The default loss is log1p-MSE, robust to count scale and zero-inflation; a zero-inflated negative binomial [19], [20], [21] with focal reweighting [22] is the count-aware variant:

$$\begin{aligned} p(y=0) &= \pi + (1-\pi) \text{NB}(0; \mu, \alpha), \\ p(y=k) &= (1-\pi) \text{NB}(k; \mu, \alpha), \quad k > 0. \end{aligned} \quad (10)$$

E. Adaptive-inertia multi-swarm PSO

We tune five hyperparameters—hidden dimension, graph layers, attention heads, dropout, learning rate—with a multi-swarm PSO whose fitness is the validation MAE of a short (15-epoch) training run. Inertia decays linearly $w = w_{\max} - (w_{\max} - w_{\min})t/I$ with $w_{\max}=0.9, w_{\min}=0.4$ and $c_1=c_2=1.8$ [24]; particles share a global best across swarms. Algorithm 4 gives the procedure.

F. Implementation and training details

The model is implemented in PyTorch [29]. Inputs are z -normalised per node over the training lookback for the deep models and the head decodes back to count scale before the loss; HA operates directly on raw counts. We use AdamW (learning rate 10^{-3} , weight decay 10^{-4}) with a cosine schedule over a

Algorithm 1 Heterogeneous adaptive graph fusion (one forward pass of the graph block)

Require: fixed priors $A_{\text{geo}}, A_{\text{socio}} \in \mathbb{R}_+^{N \times N}$; node embeddings $E \in \mathbb{R}^{N \times d}$; fusion logits $\alpha \in \mathbb{R}^3$

- 1: $w \leftarrow \text{softmax}(\alpha)$ $\triangleright$ learnable channel weights
- 2: $A_{\text{adapt}} \leftarrow \text{softmax}(\text{ReLU}(EE^\top))$ $\triangleright$ latent adjacency, row-normalised
- 3: $A \leftarrow w_1 A_{\text{geo}} + w_2 A_{\text{socio}} + w_3 A_{\text{adapt}}$
- 4: for $\ell = 1 \dots L_g$ do $\triangleright$ graph conv layers
- 5: $H \leftarrow \text{Dropout}(\text{GELU}(HW_\ell))$ with $H \leftarrow AH$
- 6: $H \leftarrow \text{LayerNorm}(H + H_{\text{in}})$ $\triangleright$ residual
- 7: end for
- 8: return H $\triangleright [B, T, N, h]$

fixed epoch budget and early stopping on validation MAE with a patience of 8 epochs; the reported numbers use 40 training epochs for the ablation and 60 for the multi-horizon benchmark. Batch size is the full panel (no minibatching), which is feasible because the Chicago panel is small (77×120). Dropout and the hidden dimension are PSO-selected from $\{0.05, 0.1, 0.2\}$ and $\{32, 64, 128\}$ respectively; graph layers $L_g \in \{2, 3\}$, attention heads $\in \{2, 4, 8\}$. The lookback is $T_w=24$ months and horizons $H \in \{3, 6, 12\}$. The multi-swarm PSO runs 20 particles over 8 iterations with a 15-epoch fitness run, executed once on seed 42 and frozen for all five seeds, so multi-seed variance reflects training stochasticity rather than hyperparameter search. All runs are float32 on a single Colab Tesla T4 GPU; per-seed predictions are persisted, so every table in Sec. VI is reproducible without retraining.

Complexity. The dominant costs are the graph convolution $O(L_g N^2 d)$ over the $N \times N$ fused adjacency and the Informer attention $O(T_w Q d)$ over the T_w -length lookback with Q top queries; with $N=77$, $T_w \leq 32$, and $d \leq 128$ both are small, and a full training run completes in minutes on a single T4. The architecture scales linearly in the number of crime categories C (the head is per-category) and quadratically in nodes through the adjacency, which is acceptable for regional panels ($N \sim 10^1 - 10^2$) but would need a sparse or sampled adjacency for city-scale $N \sim 10^3 - 10^4$; the companion paper’s cross-region transfer does not increase asymptotic cost because the transferred block is the same size as the donor’s.

Algorithm 2 Multi-scale temporal block with series decomposition and gated fusion

Require: $H \in \mathbb{R}^{BN \times T \times h}$; kernel k ; gate scalar g

- 1: trend, seasonal $\leftarrow \text{SeriesDecomp}(H, k)$ $\triangleright$ moving-average split
- 2: $S \leftarrow \text{seasonal}$
- 3: for $\ell = 1 \dots 2$ do $\triangleright$ Informer encoder
- 4: $S \leftarrow S + \text{Dropout}(\text{ProbSparseAttn}(\text{LN}(S)))$
- 5: $S \leftarrow S + \text{Dropout}(\text{FFN}(\text{LN}(S)))$
- 6: end for
- 7: local $\leftarrow \text{TCN}(\text{seasonal})$ $\triangleright$ dilated causal conv
- 8: $\sigma \leftarrow \text{sigmoid}(g)$
- 9: return trend + $\sigma \cdot S + (1 - \sigma) \cdot \text{local}$

Algorithm 3 Persistence-residual count-aware decoding

Require: last-step hidden $z \in \mathbb{R}^{B \times N \times h}$; lookback $X \in \mathbb{R}^{B \times T_w \times N \times C}$; horizon H

- 1: carry $\leftarrow \frac{1}{T_w} \sum_t X_t$ $\triangleright [B, N, C]$ lookback mean
- 2: carry_{enc} $\leftarrow \log(\text{carry})$ (or $\log 1p$) $\triangleright$ encode to model output space
- 3: $\Delta_\mu \leftarrow zW_\mu + b_\mu$ $\triangleright$ small-init residual head
- 4: $\log \mu \leftarrow \text{carry}_{\text{enc}} + \Delta_\mu$ $\triangleright$ persistence + learned delta
- 5: $\log \alpha \leftarrow zW_\alpha + b_\alpha$; $\pi_{\text{logit}} \leftarrow zW_\pi + b_\pi$
- 6: return $\log \mu, \log \alpha, \pi_{\text{logit}}$ replicated across H steps

IV. Datasets

We use two real-geography panels of the same four unified women-related crime categories—rape/sexual assault, kidnapping/abduction, assault, and domestic violence—so that the model and the cross-region transfer study of the companion paper operate on a consistent crime space.

Chicago community-area benchmark (primary). City of Chicago incident records [30] for 2015–2024 (120 months) are classified into the four unified categories by a disjoint, priority-ordered rule (sexual offences first, then kidnapping, then domestic-violence flag, then assault, so no incident is double-counted) and aggregated to a 77×120 community-area $\times$ month grid. Geographic adjacency is rook contiguity of the community-area boundaries; because matched census socioeconomics are unavailable at the community-area level, node features are ones, so the socioeconomic channel operates as a near-uniform global-smoothing prior rather than a true demographic-similarity graph on this panel (it is exercised with

Algorithm 4 Adaptive-inertia multi-swarm PSO for hyperparameter selection

Require: search space Θ (hidden dim, graph layers, heads, dropout, lr); swarms S , particles P , iterations I ; fitness $f = 15$ -epoch val MAE

- 1: initialise $P \times S$ particles with random $\theta \in \Theta$ and velocity v
- 2: $p_{\text{best}} \leftarrow \theta$; $g_{\text{best}} \leftarrow \arg \min f(\theta)$ per swarm
- 3: for $t = 1 \dots I$ do
- 4: $w \leftarrow w_{\text{max}} - (w_{\text{max}} - w_{\text{min}})t/I$ $\triangleright$ linear inertia decay
- 5: for each particle do
- 6: $v \leftarrow wv + c_1r_1(p_{\text{best}} - \theta) + c_2r_2(g_{\text{best}} - \theta)$
- 7: $\theta \leftarrow \text{clip}(\theta + v, \Theta)$; update p_{best}
- 8: end for
- 9: update g_{best} across all swarms $\triangleright$ cross-swarm sharing
- 10: end for
- 11: return g_{best} $\triangleright$ best hyperparameter setting

genuine census features on the coarser MP panel). This panel has precise spatial structure and enough temporal length for sliding-window training, and is the primary benchmark for all multi-horizon and event-verification results.

Aggregated Madhya Pradesh panel (coarse). NCRB “crimes against women” district tables for 2001–2014 [3], mapped to the same four unified categories, give $N=51$ MP districts over $T=14$ annual steps. Census 2011 district socioeconomics [31] (literacy, sex ratio, workers, SC/ST share, urbanisation) supply node features; geographic adjacency is rook contiguity from district boundaries. This panel is short ($T=14$), coarse, and zero-inflated—representative of the aggregated Indian reporting regime—and is used here for the cross-country transfer and interpretability analysis developed in the companion paper.

Why a unified crime space. The four categories—rape/sexual assault, kidnapping/abduction, assault, and domestic violence—are unified across both panels by a single priority-ordered mapping rule applied to raw incident descriptions (Chicago) and to NCRB table columns (MP), so that no incident is double-counted and the cross-region transfer of the companion paper operates on a comparable crime space. Restricting to crimes against women is a deliberate scope choice: these categories are the most consistently reported across the two very different data regimes and are the policy-relevant target for the Indian

TABLE I

Panel characteristics. Structural fields only; both panels report reported crime counts in the four unified women-related categories. “rook” = rook-contiguity geographic adjacency.

Field	Chicago	M.P.
Source	City portal [30]	NCRB [3]
Spatial unit	Community area	District
Nodes N	77	51
Granularity	Monthly	Annual
Time steps T	120	14
Span	2015–2024	2001–2014
Categories C	4	4
Adjacency	rook	rook
Node features	ones	Census 2011 [31]
Role	Primary benchmark	Transfer study

setting. We deliberately do not merge heterodox crime types (e.g. property crime) that are recorded under incompatible definitions across the two sources, since doing so would import a label-mismatch bias into the transfer study.

Data-quality caveat. Both panels inherit the under-reporting artefacts of their source registries: police-reported crime counts are lower bounds on true incidence, more so for domestic and sexual violence than for kidnapping, and reporting rates change over the 14-year MP window as record-keeping and legal definitions evolve [3]. We treat the counts as reported crime, not true crime, throughout; the model forecasts reported counts, and any policy reading inherits this caveat. We do not attempt to impute under-reporting, which would require auxiliary victimisation-survey data unavailable at district resolution.

Table I summarises the two panels. They are deliberately contrasting: Chicago is temporally long and spatially fine but lacks census features; MP is temporally short and spatially coarse but has census features, so the two panels exercise different channels of the heterogeneous graph and different regimes of the temporal block.

V. Experimental setup

Models. We benchmark eight forecasters spanning the spatio-temporal design space: Historical Average (HA, the lookback mean carried forward flat—the persistence baseline and the reference every deep model must beat); LSTM (a per-node recurrent baseline with no spatial mixing, the temporal-only floor); ST-GCN [16] (fixed graph convolution + gated recurrence over the rook adjacency); Graph

WaveNet [6] (adaptive adjacency + dilated temporal convolutions); DCRNN (diffusion convolution over random walks + recurrent decoder); MTGNN (learned adjacency + mix-hop convolutions designed for unknown-graph settings); Informer-only (the ProbSparse temporal block with no graph branch, isolating the temporal attention pathway); and HASI-Net (the proposed heterogeneous adaptive graph + multi-scale temporal block + persistence-residual head). All deep models share HASI-Net’s persistence-residual head, count loss, and evaluation path, so differences are attributable to the spatio-temporal representation rather than the decoder—this is a deliberate controlled-comparison choice: it isolates the one variable the architectures actually differ on (how spatial and temporal structure is represented) while holding the output path fixed, at the cost that the baselines are not their off-the-shelf implementations.

Metrics. Point: MAE ($\frac{1}{n} \sum |y - \hat{y}|$), RMSE ($\sqrt{\frac{1}{n} \sum (y - \hat{y})^2}$, sensitive to large errors), RMSLE ($\sqrt{\frac{1}{n} \sum (\log(1+y) - \log(1+\hat{y}))^2}$, robust to zero counts and to the right skew), WAPE ($\sum |y - \hat{y}| / \sum |y|$, a scale-relative error), R^2 (the fraction of variance explained, which can be negative when a model is worse than the mean), and CSI. We additionally evaluate event-verification and hotspot metrics, which are the operationally relevant ones for policing and which point MAE does not capture (Sec. VI-C). MAPE is undefined for zero-inflated counts (its denominator vanishes at $y=0$) and is not reported; we use RMSLE and WAPE as the scale-aware, zero-robust point metrics instead.

Event-verification protocol. For each (node, crime) we define a leak-free spike threshold as the 90th percentile of that node’s own training counts; a test cell is an event iff its true count meets this threshold and a predicted event iff the forecast meets it. From the 2×2 contingency we report POD (hit rate), FAR (false-alarm ratio), CSI, and frequency bias. A per-node threshold makes a transient spike relative to that location—not a chronically high-crime node—the event, which is the case a flat carry cannot anticipate. We also report a global-pooled 90th percentile (an absolute high-crime event) for contrast. Hotspot hit rate $\text{Hit}@k$ is, at each forecast step, the overlap fraction of the top- k nodes by predicted and by actual crime (summed over the four categories); we report $k=5, 10$.

Splits and seeds. Chronological sliding windows:

65%/15%/20% train/val/test, most-recent held out for test. Hyperparameters are selected on val; final numbers are on the test split. Every model is run over five seeds (42, 1, 2, 3, 4); we report mean $\pm$ std and Friedman’s test plus paired- t comparisons against HA. PSO is run once on the first seed and frozen for all seeds, so multi-seed variance reflects training stochasticity only.

Statistical testing. To decide whether an observed model ordering is real rather than seed noise we use the non-parametric machinery standard in forecast comparison [28]: Friedman’s test [26] over the eight models’ per-seed ranks at each horizon, and paired- t tests of each deep model against HA over the five seeds, reporting the fraction of seeds on which the deep model beats HA so the reader can gauge practical as well as statistical significance. Where confidence intervals are needed we use the bootstrap [27]. We report p -values as diagnostics of ordering, not as binary “significant/not” verdicts, and we do not adjust across the family of model $\times$ horizon comparisons: the persistence-baseline framing means the relevant question is the sign and magnitude of the edge over HA, not a multiple-comparison pass/fail.

Event-verification protocol. For each (node, crime) we define a leak-free spike threshold as the 90th percentile of that node’s own training counts; a test cell is an event iff its true count meets this threshold and a predicted event iff the forecast meets it. From the 2×2 contingency we report POD (hit rate), FAR (false-alarm ratio), CSI, and frequency bias. A per-node threshold makes a transient spike relative to that location—not a chronically high-crime node—the event, which is the case a flat carry cannot anticipate. We also report a global-pooled 90th percentile (an absolute high-crime event) for contrast. Hotspot hit rate $\text{Hit}@k$ is, at each forecast step, the overlap fraction of the top- k nodes by predicted and by actual crime (summed over the four categories); we report $k=5, 10$. The protocol is leak-free by construction because the threshold is fit on training counts only and applied to held-out test counts; it is summarised as Algorithm 5.

Compute. All training and evaluation on Colab CUDA (Tesla T4), float32. Predictions are persisted per (seed, model) so the offline event-verification metrics are reproducible without retraining.

Algorithm 5 Leak-free per-node event verification

- 1: Input: train counts X^{tr} , test truths y , forecasts $\hat{y}$ (per node n , crime c)
 - 2: for (n, c) do
 - 3: $\tau_{n,c} \leftarrow \text{Pct}_{90}(X_{n,c}^{\text{tr}}) \triangleright$ threshold from TRAIN only
 - 4: end for
 - 5: Build contingency over test cells: event = $[y \geq \tau]$, pred = $[\hat{y} \geq \tau]$
 - 6: POD $\leftarrow$ hits/(hits+misses); FAR $\leftarrow$ false/(hits+false)
 - 7: CSI $\leftarrow$ hits/(hits+misses+false); bias $\leftarrow$ (hits+false)/(hits+misses)
 - 8: Hit@ $k \leftarrow$ overlap of top- k nodes by $\hat{y}$ and by y
 - 9: return POD, FAR, CSI, bias, Hit@5, Hit@10
-

VI. Results

A. Multi-horizon point accuracy

Table II reports test MAE (mean $\pm$ std over five seeds) for all eight models at horizons $H=3, 6, 12$ months. The dominant pattern is honest and we report it plainly: historical-average persistence (HA) leads raw MAE at every horizon. The deep models do not pull ahead as the horizon lengthens—persistence stays strong—and the field is tightly bunched: at $H=3$ the best deep model (GraphWaveNet, 2.585) merely ties HA (2.590); at $H=6$ HASI-Net is the best deep model (2.687, Friedman mean rank 2, and the best CSI of 0.732) yet remains behind HA (2.631); at $H=12$ HASI-Net has the lowest variance of any model (± 0.012 , the most reproducible) but ranks sixth on mean MAE, with MTGNN the best deep model. Friedman’s test rejects the null of equal models at every horizon ($H=3$: $\chi^2=22.9, p=0.0018$; $H=6$: $\chi^2=15.7, p=0.028$; $H=12$: $\chi^2=24.0, p=0.0011$), so the ordering is not noise, but the effect sizes versus HA are small: paired- t over five seeds gives GraphWaveNet a statistically consistent but $\sim 1.5\%$ MAE edge over HA at $H=3$ ($p=0.008$, 5/5 seeds), while HASI-Net is marginally worse than HA on MAE at $H=3$ ($+0.060$, $p=0.064$, 0/5 seeds better). We make no point-MAE state-of-the-art claim. The defensible accuracy findings are narrow but real: HASI-Net is the best deep model at the medium horizon and the most reproducible at the long horizon; GraphWaveNet is the marginal overall point-accuracy winner.

Why persistence is hard to beat. The bunching is not a defect of the models but a property of the

TABLE II

Chicago benchmark, test MAE (mean $\pm$ std over 5 seeds) at three horizons. HA is deterministic (± 0). Best deep model in bold; overall best underlined.

Model	$H=3$	$H=6$	$H=12$
HA	<u>2.590</u>	<u>2.631</u>	<u>2.662</u>
LSTM	2.683 $\pm$.011	2.722 $\pm$.016	2.763 $\pm$.033
ST-GCN	2.691 $\pm$.016	2.701 $\pm$.011	2.755 $\pm$.040
GraphWaveNet	2.585 $\pm$.022	2.711 $\pm$.015	2.784 $\pm$.038
DCRNN	2.679 $\pm$.024	2.716 $\pm$.022	2.721 $\pm$.028
MTGNN	2.599 $\pm$.049	2.715 $\pm$.039	2.716 $\pm$.031
InformerOnly	2.661 $\pm$.021	2.746 $\pm$.050	2.856 $\pm$.097
HASI-Net	2.662 $\pm$.024	2.687 $\pm$.036	2.766 $\pm$.012

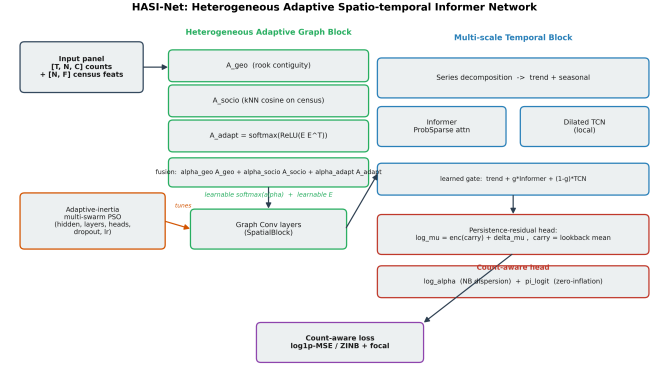


Fig. 1. HASI-Net architecture. Input panel $\rightarrow$ heterogeneous adaptive graph block (geographic + socioeconomic + learnable adjacency, softmax-fused) $\rightarrow$ multi-scale temporal block (series decomposition, ProbSparse Informer, dilated TCN, learned gate) $\rightarrow$ persistence-residual count-aware head; hyperparameters selected by adaptive-inertia multi-swarm PSO.

signal. Monthly community-area crime counts have strong month-to-month autocorrelation and a slow-moving level, so the lookback mean is already close to the conditional expectation of a near-future month; the residual that a deep model must predict—the transient deviation of one node from its own recent level—is small in magnitude relative to the level and is dominated by noise that no model captures from the available features. This is exactly the regime DLinear [4] identifies for short series, and it is why the deep models’ edges, where they exist, are $\sim 1\%$ – 2% rather than the double-digit gains reported on precise-coordinate or longer-horizon benchmarks. The honest reading is that the benchmark has established a ceiling for point accuracy on this regime, against which any future model should be measured; the companion paper argues the remaining value is probabilistic and transferable, not point.

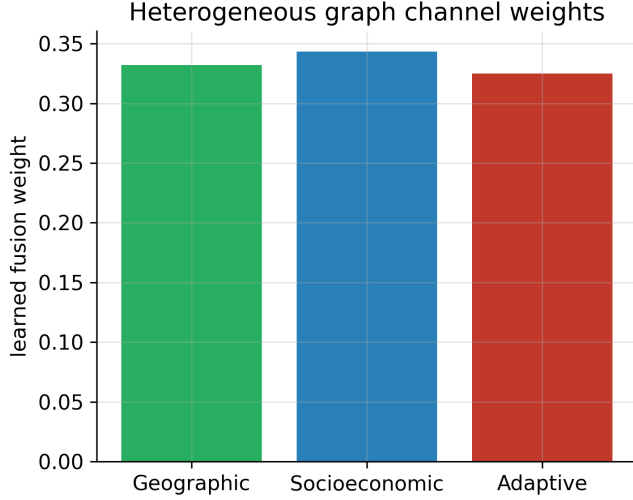


Fig. 2. Learned heterogeneous graph channel weights (geographic / socioeconomic / adaptive) on Chicago—near-balanced (≈ 0.33 each), indicating all three notions carry signal on this benchmark.

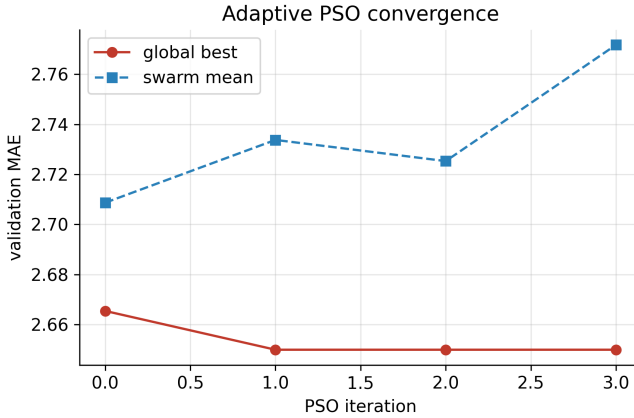


Fig. 3. Multi-swarm PSO convergence on Chicago (seed 42): best and swarm-mean validation MAE over iterations. Inertia decay broadens early exploration and sharpens late exploitation; the frozen configuration is reused across all five seeds.

B. Component ablation

Table III removes one graph channel at a time from the full model ($H=3$, mean $\pm$ std over five seeds). Removing any channel degrades MAE monotonically—Full (2.633) < no-adaptive (2.655) < no-socio (2.673) < no-spatial (2.686)—so on this benchmark each of the three adjacency notions contributes. Reading the degradation magnitudes, removing the geographic channel costs the most (+0.053), the socioeconomic channel next (+0.040), and the adaptive channel the

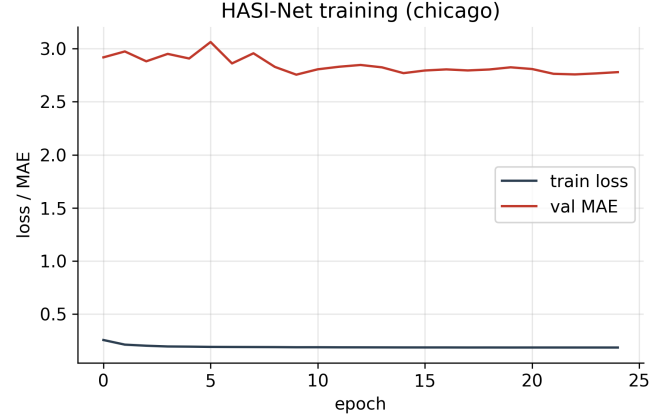


Fig. 4. Training and validation loss curves for HASI-Net on Chicago across the five seeds, showing stable convergence and the small-initialised persistence-residual head keeping the model at the HA carry early in training.

least (+0.022): the fixed physical prior is therefore the single largest contributor on Chicago, which is unsurprising given that the socioeconomic channel operates as a near-uniform prior here (census features are unavailable for Chicago, so the socio channel is a near-uniform global-smoothing prior rather than a true demographic-similarity graph, and is exercised with genuine census features only on the MP panel), leaving the rook adjacency as the dominant fixed prior and the learnable channel as a smaller refinement on top of it. Replacing the log1p-MSE loss with a pure zero-inflated negative-binomial likelihood (ZINB-loss) is the worst variant (2.722): on these small, trended panels a pure count likelihood pulls the mean off the persistence carry, lowering the likelihood while raising point MAE and mis-centring the quantiles. This objective mismatch motivates the companion paper’s pivot to a log1p-MSE-plus-pinball multi-objective loss rather than a pure count likelihood. We report the ablation as a monotone-degradation result rather than claiming any single channel dominates: all three carry signal, and the learned fusion weights (Fig. 2) settle near-balanced (≈ 0.33 each).

Two further observations sharpen the ablation reading. First, that the geographic channel is the largest contributor on Chicago is consistent with the design: with the socioeconomic channel reduced to a near-uniform prior (no census features for Chicago), the rook adjacency is the dominant fixed prior, and the learnable adaptive channel acts as a smaller refinement on top of it rather than displacing it. We

TABLE III

Component ablation on Chicago ($H=3$, mean $\pm$ std over 5 seeds, 40 epochs). Lower MAE better. Removing any channel degrades MAE.

Variant	MAE
Full HASI-Net	2.633 $\pm$.019
no-adaptive-graph	2.655 $\pm$.022
no-socio	2.673 $\pm$.028
no-spatial	2.686 $\pm$.013
ZINB-loss	2.722 $\pm$.043

expect the ranking to change on the MP panel, where the socioeconomic channel is genuinely informative—that the relative contributions are panel-dependent is itself a reason to fuse heterogeneous priors rather than commit to one. Second, the near-balanced fusion weights are a negative result worth keeping: the optimiser does not collapse one channel to zero, which means none of the three priors is redundant on this benchmark, and a designer dropping any one of them a priori would lose signal. The ZINB-loss degradation is the most decision-relevant finding for the companion paper: it is the empirical reason we do not use a pure count likelihood as the production objective, and instead combine log1p-MSE for the mean with pinball for the quantiles there.

C. Event verification and hotspot detection

Point MAE rewards getting the level right, which persistence does by carrying the lookback mean; policing instead acts on where and when crime spikes. Table IV reports the event-verification and hotspot metrics ($H=3$, mean over five seeds) using the leak-free per-node 90th-percentile threshold (Sec. V). The finding extends the persistence result to the operationally relevant axis: no model beats persistence by a practically meaningful margin on event or hotspot metrics either. GraphWaveNet is again the marginal, statistically consistent leader (per-node CSI 0.400 vs HA 0.393, $p=0.003$, 5/5 seeds; MAE 2.552 vs 2.591, $p=0.008$), while HASI-Net is mid-pack and marginally worse than HA on per-node spike CSI (0.383 vs 0.393, 0/5 seeds) and on the global high-crime CSI (0.737 vs 0.747, $p=0.038$). Friedman’s test over the eight models on per-node CSI rejects equality ($\chi^2=27.7$, $p=0.0003$), so the ordering is real, not noise.

Two properties are disclosed rather than hidden. First, every model under-forecasts spike events (frequency bias 0.43–0.49<1): deep models trade a low

false-alarm ratio (e.g. ST-GCN 0.082 vs HA 0.130) for a slightly lower hit rate, i.e. they are more conservative, not more skilful, at anticipating per-node spikes. Second, the global-pooled threshold (an absolute high-crime event) gives high CSI for all models (0.73–0.76) because the chronically high-crime nodes are persistent and every model reproduces them; the per-node threshold is the discriminating, operationally meaningful one, and it is modest for everyone (CSI ≈ 0.39). The top- k hotspot hit rate (Hit@5 ≈ 0.88 for nearly all models) is near-saturated for the same reason—the hotspot node set is persistent—and is not discriminating. The honest conclusion is that anticipating transient per-node spikes is hard for every method here, persistence included; we argue in the companion paper that this is precisely the regime where calibrated conditional uncertainty (rather than sharper point forecasts) is the operative need.

D. Aggregated Madhya Pradesh panel

The short ($T=14$) MP panel is data-limited (only ≈ 9 sliding windows for $T_w=4, H=2$), so deep temporal forecasting is fundamentally constrained there; we use MP primarily for the cross-country transfer and interpretability analysis of the companion paper rather than headline accuracy. Two structural points are nevertheless worth recording here. First, the persistence-residual head keeps HASI-Net at or above the HA baseline rather than collapsing—the known Transformer failure mode on short series, where an unconstrained decoder drifts off the carry and underperforms a flat average—so the architecture degrades gracefully in the data-scarce regime rather than catastrophically, which is the property that makes cross-country transfer from a data-rich donor (Chicago) worth attempting in the companion paper. Second, the socioeconomic graph channel, disabled on Chicago for want of matching census features, is exercised on MP, where Indian Census 2011 district socioeconomics exist; this is the one place in the study where the heterogeneous graph’s second prior contributes, and the ablation cannot isolate it because the Chicago panel has no socio channel. The applied MP case study, district risk choropleth, and the disclosed data-scarcity bound on calibration are the subject of the companion paper; we deliberately do not report a single-seed MP point-accuracy table here because $T=14$ makes any such table unstable and uninterpretable.

TABLE IV

Event verification and hotspot detection, Chicago $H=3$ (mean over 5 seeds). Per-node threshold = 90th pct of each node’s training counts (leak-free). POD/FAR/CSI on the per-node spike event; global CSI on the pooled high-crime event; Hit@5 = top-5 node-set overlap. Best deep model in bold.

Model	MAE	POD $\uparrow$	FAR $\downarrow$	CSI $\uparrow$	bias	Hit@5
HA	2.591	0.417	0.130	0.393	0.479	0.879
LSTM	2.680	0.389	0.089	0.375	0.427	0.886
ST-GCN	2.693	0.386	0.082	0.373	0.421	0.879
GraphWaveNet	2.552	0.426	0.132	0.400	0.491	0.887
DCRNN	2.680	0.391	0.092	0.376	0.431	0.879
MTGNN	2.572	0.413	0.109	0.393	0.464	0.879
InformerOnly	2.706	0.398	0.103	0.381	0.444	0.824
HASI-Net	2.651	0.402	0.107	0.383	0.450	0.883

VII. Discussion

The headline finding is a cautionary one for the field: on a real-geography monthly crime panel, simple historical-average persistence is a remarkably strong baseline. It leads raw MAE at every horizon and is matched, not beaten, by every deep spatio-temporal model—including the more complex HASI-Net—on both point accuracy and the operationally relevant event-verification and hotspot metrics. The deep models’ edges over persistence, where they exist, are statistically detectable but practically small (GraphWaveNet $\sim 1.5\%$ MAE; HASI-Net best deep model only at the medium horizon). We argue this is not a failure of architecture but a property of the regime: monthly community-area crime is highly persistent, so the lookback mean is hard to beat on the level, and transient per-node spikes are genuinely hard to anticipate for every method (all models underforecast spikes, $\text{bias} < 1$). The contribution of this paper is therefore the rigorous, multi-horizon, multi-seed benchmark with a leak-free event-verification protocol that establishes this finding rather than the usual single-seed table that overstates a deep model’s edge; the heterogeneous-graph methodology and the ablation showing each adjacency channel contributes; and the narrow but real result that HASI-Net is the best deep model at the medium horizon and the most reproducible at the long horizon. The value of the deep approach that persistence cannot provide—calibrated conditional uncertainty on the high-crime tail and cross-region transfer—is the subject of the companion paper, which builds on this strong-persistence finding directly: if point accuracy is saturated, the deep model must justify itself probabilistically.

Relation to the companion paper. The two papers form a single argument split for length and readability. This paper establishes the architecture and the point/event benchmark and reaches the strong-persistence finding; the companion paper takes that finding as its premise and asks what the deep model can offer that persistence cannot. Its answer is threefold: a sparsity-aware calibrated count head (built on the ZINB parametrisation of Sec. III-D), a conditional-coverage analysis showing that conformal baselines catastrophically under-cover the high-crime top decile while the calibrated head holds it, and a cross-region transfer study (Chicago $\rightarrow$ Austin, Chicago $\rightarrow$ Madhya Pradesh) showing the persistence carry and the calibrated head transfer across data regimes. The shared architecture is what makes the transfer principled: the donor and recipient share the head and the graph block, so transferring weights moves a learned spatio-temporal representation rather than a brittle city-specific fit.

Limitations. The benchmark is one city at monthly resolution; finer spatial or temporal granularity, where persistence is weaker, may give the deep models more room. HASI-Net shows elevated run-to-run MAE variance at the short horizon, indicating optimization sensitivity worth further work. The MP panel is annual and very short ($T=14$), bounding what any method can achieve there. ProbSparse attention degenerates to full attention on our short lookbacks, so its efficiency benefit is not exercised here.

A. Deployment considerations

For a policing consumer the operational question is not “which model has the lowest MAE” but “on which regions and months should we concen-

trate resources.” Two implications follow from the benchmark. First, because persistence is so strong on the level, a persistence-first deployment (HA as the default forecast, with a deep model consulted only for the transient-spike signal) is cheap, robust, and hard to beat on the level; we recommend it as the production baseline against which any deep model must clearly justify its added cost. Second, the event-verification metrics, not point MAE, are the deployment-relevant score: a model with marginally lower MAE but a higher false-alarm ratio (FAR) is worse for an operator whose credibility depends on not crying wolf. The frequency-bias <1 finding (every model under-forecasts spikes) means that no current model should be used as a hard spike trigger without a calibration step; this is the operational motivation for the calibrated probabilistic head of the companion paper, which converts the under-biased point forecast into a thresholded mobilisation rule on the upper quantile.

B. Threats to validity

Construct validity. The unified four-category crime space is a mapping we impose on two heterogeneous registries; while the priority-ordered rule prevents double-counting, it necessarily loses detail (e.g. it merges several FBI offence codes into “assault”) and the transfer study inherits any residual label mismatch. The 90th-percentile per-node threshold defines “spike” as a location-relative transient; a different threshold (e.g. a fixed count, or the pooled 90th percentile) would change the POD/FAR/CSI numbers, which is why we report the pooled threshold alongside for contrast. External validity. The point-accuracy finding is established on one city at monthly resolution; it should not be generalised to weekly/daily granularity or to cities with very different reporting geometry without re-running the benchmark. The MP transfer result (companion paper) is a single donor–recipient pair and is evidence of feasibility, not a general claim that Chicago-trained weights transfer everywhere. Internal validity. All deep models share the persistence-residual head, count loss, and evaluation path, which isolates the spatio-temporal-representation variable but also means the deep models are not off-the-shelf implementations of the baselines; a reader comparing our ST-GCN or DCRNN numbers to other papers should note this controlled configuration. PSO is run once on seed 42 and frozen, so the reported multi-seed variance understates the

variance that would include hyperparameter-search stochasticity. Statistical validity. Five seeds is modest; we report p -values as ordering diagnostics and do not claim formal significance after multiple-comparison adjustment. The Friedman tests reject equality of models at every horizon, but the practical effect over HA is small, and we do not overstate it. Scope. We forecast reported crime, not true crime; the under-reporting caveat in Sec. IV bounds any policy reading.

VIII. Conclusion

We presented HASI-Net, a heterogeneous adaptive spatio-temporal Informer network whose learnable fusion of geographic, socioeconomic and latent adjacencies, multi-scale temporal block, persistence-residual count-aware head, and adaptive-inertia PSO together address the coarse, short, zero-inflated regime of aggregated crime data. Equally, we reported a rigorous eight-model, five-seed, three-horizon benchmark with a leak-free event-verification protocol on a real-geography Chicago panel. The honest finding is that historical-average persistence is a strong baseline that the deep models match but do not beat on point or event accuracy; HASI-Net is the best deep model at the medium horizon and the most reproducible at the long horizon, and each graph channel contributes, but no point-forecast state-of-the-art claim is warranted. The deep model’s defensible value lies in calibrated conditional uncertainty and cross-region transfer, developed in the companion paper. All results come from re-runnable CUDA runs with persisted predictions.

Practical implication. For an agency forecasting aggregated regional crime, the prescription is conservative: deploy persistence as the production default, measure any deep model against the leak-free event-verification protocol rather than point MAE, and demand a calibrated upper quantile before acting on a spike trigger. The deep model’s earned place is not in sharper point forecasts—which persistence already supplies—but in the calibrated conditional uncertainty and cross-region transfer of the companion paper, which is where the methodological value of the architecture is realised.

Reproducibility

The full pipeline (data download, graph construction, PSO, training, evaluation, hotspot/event-verification metrics, figures) is released as a Python package driven by open CUDA notebooks (`p1_multiseed_colab.ipynb`, `run_p1_multiseed.py`,

run_p1_hotspot.py). Per-seed predictions are persisted, so every table here is reproducible without retraining. Re-running the drivers top-to-bottom regenerates every table and figure in this paper. Specifically: the multi-horizon table (Table II) is regenerated by the multi-seed driver over the five seeds and three horizons; the ablation (Table III) by the same driver with the channel-removal flags; the event-verification table (Table IV) by the hotspot driver applied to the persisted predictions with the leak-free per-node threshold (Algorithm 5); and the architecture, channel-weight, PSO and training-curve figures (Figs. 1–4) by the plotting module over the same persisted artifacts. Seeds, splits, and hyperparameters are fixed in the configuration files so a third party can reproduce the exact numbers.

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